

EXPERIMENT – 24

Implementation of RSA Algorithm and Finding the Private Key

CODE:

```
#include <stdio.h>

int gcd(int a, int b)
{
    if (b == 0)
        return a;
    return gcd(b, a % b);
}

int main()
{
    int e = 31, n = 3599;
    int p = 0, q = 0, phi, d;

    // Find p and q
    for (int i = 2; i * i <= n; i++)
    {
        if (n % i == 0)
        {
            p = i;
            q = n / i;
            break;
        }
    }
}
```

```

    phi = (p - 1) * (q - 1);

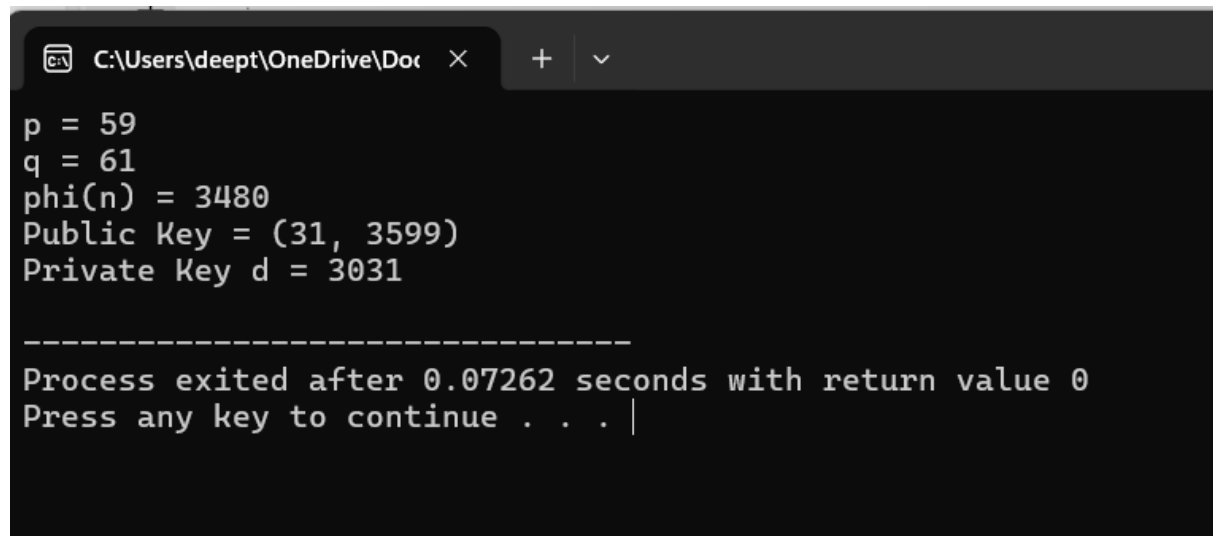
    // Find private key d
    for (d = 1; d < phi; d++)
    {
        if ((e * d) % phi == 1)
            break;
    }

    printf("p = %d\n", p);
    printf("q = %d\n", q);
    printf("phi(n) = %d\n", phi);
    printf("Public Key = (%d, %d)\n", e, n);
    printf("Private Key d = %d\n", d);

    return 0;
}

```

OUTPUT:



```

C:\Users\deept\OneDrive\Doc >
p = 59
q = 61
phi(n) = 3480
Public Key = (31, 3599)
Private Key d = 3031

-----
Process exited after 0.07262 seconds with return value 0
Press any key to continue . . .

```